# GENERATIVE AI ANALYSIS AND ETHICS REPORT

## Transformer-Based Generation of Public-Sector Contract Language

Capstone Project 5 - Generative AI Applications

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July 2026

*Research and demonstration only. Generated language is not legally valid and is not approved for legal, procurement, compliance, or operational use.*

# Executive Summary

This project implemented a custom character-level, decoder-only causal Transformer in PyTorch to generate text resembling Federal Acquisition Regulation (FAR) Subpart 52.2 provisions and clauses. The training corpus was derived from an official FAC 2026-01 FAR snapshot, validated at the source boundary, and divided at the complete-clause level into 488 training, 61 validation, and 61 held-out test records. Two otherwise identical models were trained with dropout values of 0.10 and 0.30; the 0.10 model was selected based only on validation performance. It achieved a held-out test loss of 1.141745, perplexity of 3.132, and next-character accuracy of 66.47%.

Controlled generation showed that lower predictive loss did not make the model legally reliable. Greedy decoding produced extensive repetition and weak prompt differentiation; conservative top-k sampling produced the best research-oriented balance of structure and diversity but fabricated authoritative-sounding details; and exploratory top-k sampling reduced repetition while increasing malformed words and broken syntax. The project therefore concludes that the model is suitable only for reproducible research and demonstration under strict human review.

## Project Results at a Glance

| Area | Result |
|---|---|
| Source records | 610 complete FAR provisions and clauses |
| Model | Custom character-level causal Transformer |
| Parameters | 838,400 trainable parameters |
| Selected configuration | Dropout 0.10 |
| Best validation loss | 1.125494 |
| Held-out test loss | 1.141745 |
| Held-out perplexity | 3.132 |
| Held-out token accuracy | 66.47% |
| Controlled generations | 9 outputs |
| Recommended research decoding | Conservative top-k |
| Approved for legal or operational use | No |

# 1. Overview

The generative task was to produce short continuations that resemble the surface structure of public-sector contract provisions and clauses. A Transformer was appropriate because self-attention supports sequence modeling while causal masking prevents a token position from accessing future targets during next-token prediction (Vaswani et al., 2017). The model was implemented from first principles in PyTorch rather than adapted from a pretrained language model, so the architecture, training behavior, data flow, and generation controls remain transparent and directly attributable to this project.

The task was intentionally framed as research rather than automated drafting. The FAR is an authoritative regulatory source governing federal acquisition, and the frozen project source reflects FAC 2026-01, effective March 13, 2026 (Acquisition.gov, 2026). This authoritative context makes errors especially consequential: language that looks plausible can still contain incorrect citations, dates, obligations, thresholds, or clause relationships. Accordingly, the primary analytical goal was not to demonstrate deployable legal drafting, but to evaluate how model configuration and decoding choices affect predictive performance, repetition, diversity, structural resemblance, and responsible-use risk.

## 2. Dataset or Prompt Description

The dataset was derived from the official Acquisition.gov FAR publication associated with FAC 2026-01. The extraction process isolated Subpart 52.2, which contains solicitation provisions and contract clauses, and corrected a source-boundary issue that initially allowed Subpart 52.3 material to spill into the final record. The corrected corpus ends at clause 52.253-1, contains no empty records, duplicate clause numbers, duplicate clause texts, or Subpart 52.3 spillover, and retains the source artifacts and checksums needed for auditability.

| Dataset measure | Value |
|---|---|
| Complete provisions and clauses | 610 |
| Record-text characters | 2,207,308 |
| Model-corpus characters | 2,214,359 |
| Record-text word-like tokens | 342,400 |
| Model-corpus word-like tokens | 343,620 |
| Unique word-like tokens | 8,513 |
| Empty records | 0 |
| Duplicate clause numbers | 0 |
| Duplicate text records | 0 |
| Subpart 52.3 spillover records | 0 |

Each model record prepended the official clause number to the stripped clause text. Splitting occurred before sequence-window construction and at the complete-clause level, preventing the same clause from appearing in multiple partitions. The character vocabulary was built from training records only, with validation- or test-only characters mapped to the unknown token. Sequence windows never crossed clause boundaries.

| Partition | Records | Sequence windows |
|---|---|---|
| Training | 488 | 11,489 |
| Validation | 61 | 1,408 |
| Held-out test | 61 | 1,444 |

The controlled generation experiment used three fixed FAR-style prompts: `52.212-4 `, `52.204-21 `, and `52.232-33 `. Each prompt was evaluated with greedy decoding, conservative top-k sampling, and exploratory top-k sampling using random seed 42. Every continuation was limited to 360 generated characters, yielding nine controlled outputs.

# 3. Model Design and Training Approach

The project used a decoder-only causal Transformer composed of learned character embeddings, positional embeddings, four Transformer blocks, multi-head self-attention, feed-forward networks, residual connections, layer normalization, dropout, and an output projection over the character vocabulary. The design follows the core attention-based sequence modeling principles introduced by Vaswani et al. (2017), adapted to autoregressive next-character prediction.

| Component | Configuration |
| --- | --- |
| Vocabulary size | 96 |
| Context length | 256 characters |
| Embedding dimension | 128 |
| Attention heads | 4 |
| Transformer layers | 4 |
| Feed-forward dimension | 512 |
| Trainable parameters | 838,400 |
| Objective | Next-character cross-entropy |
| Optimizer | AdamW |
| Learning rate | 0.0003 |
| Weight decay | 0.01 |
| Maximum epochs | 15 |
| Random seed | 42 |

Training used AdamW, whose decoupled weight-decay behavior is implemented directly in PyTorch (PyTorch, 2026). Padding targets were ignored by the cross-entropy loss. Before formal training, smoke tests verified tensor shapes, finite logits, strict causal invariance, gradient flow, and short optimization behavior. The formal experiment held the data, architecture, initialization controls, batch construction, optimizer settings, and epoch limit constant while changing only dropout.

## 3.1 Controlled Dropout Comparison

| Configuration | Dropout | Best epoch | Validation loss | Validation perplexity | Runtime | Peak GPU |
| --- | --- | --- | --- | --- | --- | --- |
| Baseline | 0.10 | 15 | 1.125494 | 3.082 | 133.07 s | 0.587 GB |
| Experimental | 0.30 | 15 | 1.582670 | 4.868 | 137.61 s | 0.594 GB |

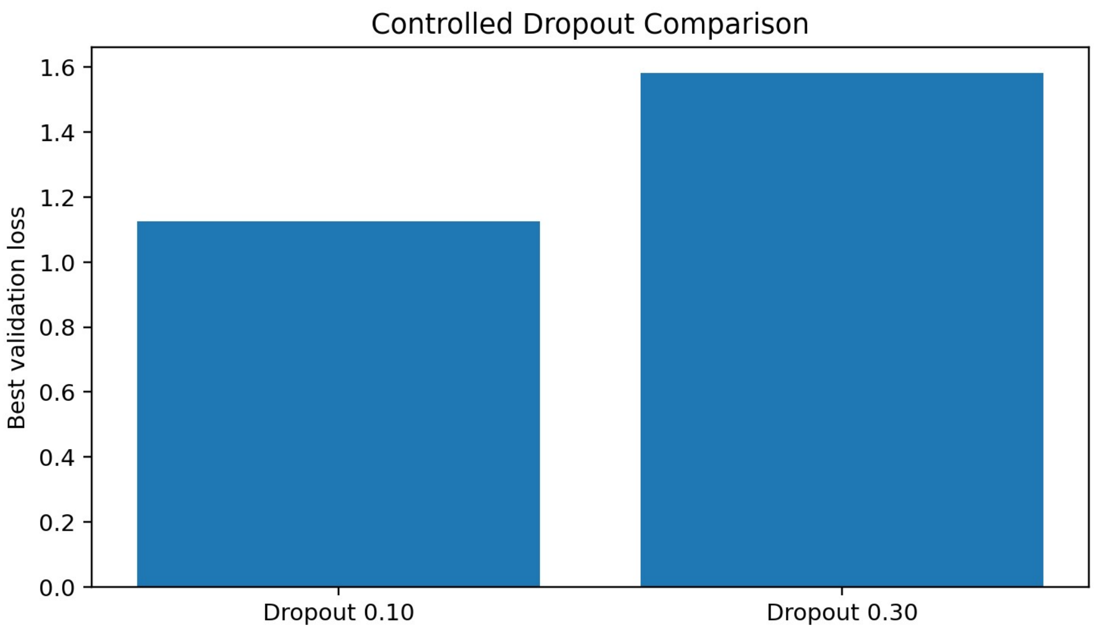


*Figure 1. Best validation loss for the controlled dropout experiment.*

The dropout 0.10 configuration achieved 28.89% lower validation loss and 36.69% lower validation perplexity with nearly identical runtime and peak GPU use. Because model selection was based only on validation performance, the held-out test partition remained untouched until after the selected inference checkpoint had been frozen and restored.

# 4. Output Evaluation and Interpretation

## 4.1 Held-Out Predictive Evaluation

| Metric | Result |
|---|---|
| Test records | 61 |
| Test sequence windows | 1,444 |
| Valid target tokens | 354,877 |
| Unknown target tokens | 15 |
| Unknown target rate | 0.0042% |
| Validation loss | 1.125494 |
| Held-out test loss | 1.141745 |
| Test minus validation loss | +0.016252 |
| Held-out test perplexity | 3.132 |
| Held-out token accuracy | 66.47% |

The small increase from validation loss to held-out test loss indicates that the validation-selected checkpoint generalized consistently to the untouched records. However, these metrics describe next-character prediction, not legal validity, factual correctness, or the quality of complete contract provisions.

The generation experiment was therefore analyzed separately using repetition, cross-prompt similarity, vocabulary diversity, sentence-terminal counts, and structured human review.

## 4.2 Controlled Decoding Results

| Setting | Temperature | Top-k | Four-gram repetition | Pairwise similarity | Human review |
|---|---|---|---|---|---|
| Greedy | 1.00 | 1 | 0.631186 | 0.768519 | 1.2 / 4 |
| Conservative top-k | 0.70 | 10 | 0.324930 | 0.073148 | 2.2 / 4 |
| Exploratory top-k | 1.00 | 25 | 0.183940 | 0.074074 | 1.8 / 4 |

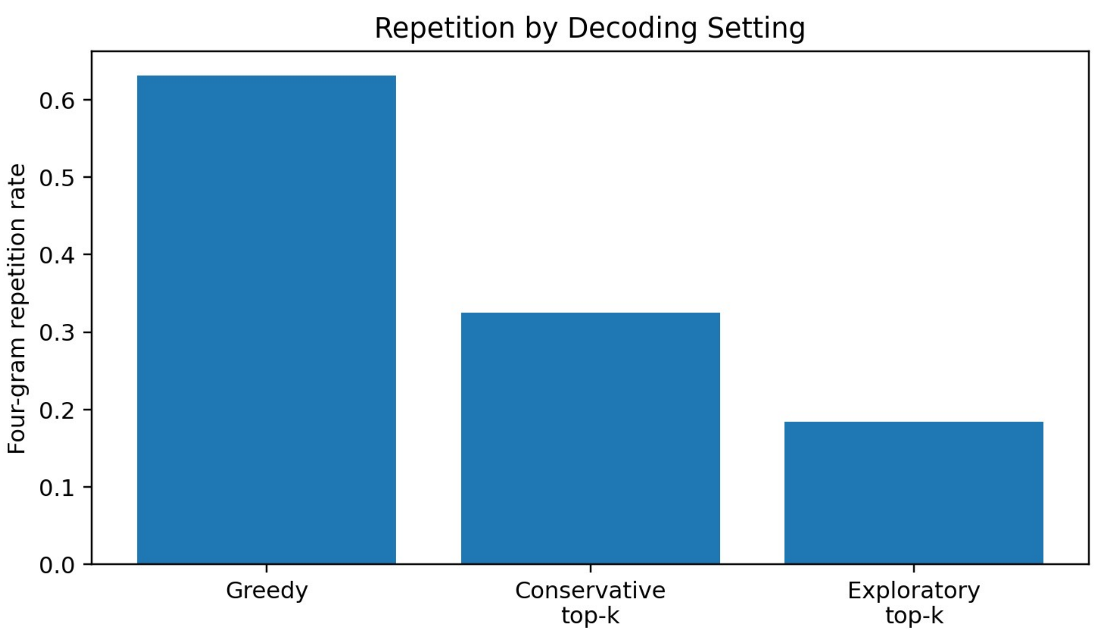


*Figure 2. Four-gram repetition rate by decoding setting.*

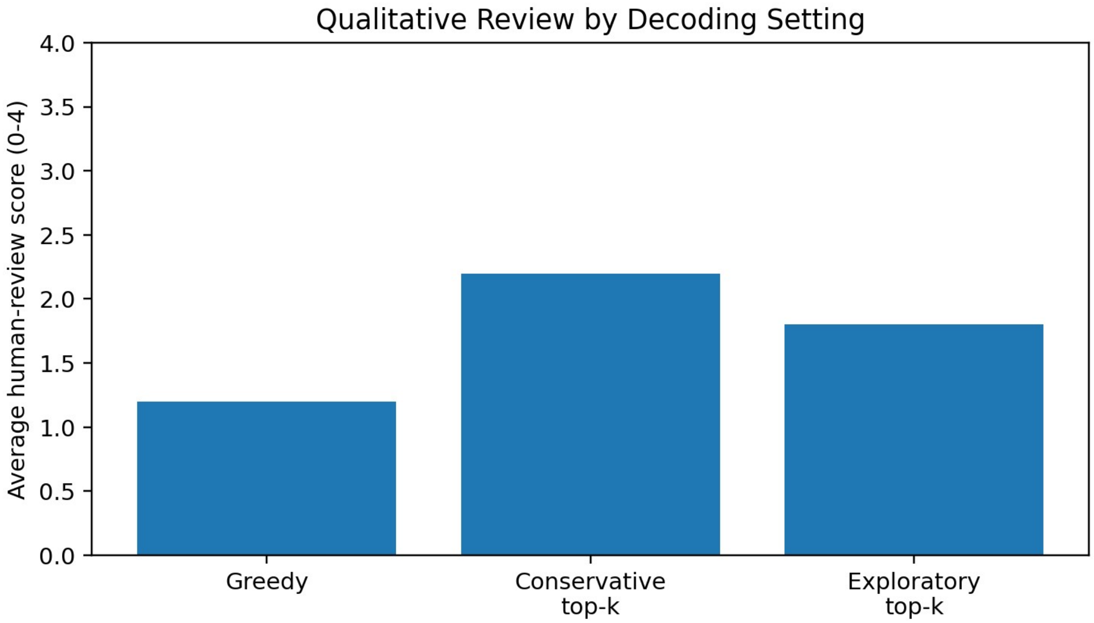


*Figure 3. Average structured human-review score by decoding setting.*

The results align with prior research showing that maximum-likelihood language models can produce bland or repetitive text under deterministic decoding and that decoding strategy can substantially alter perceived output quality without changing the underlying model (Holtzman et al., 2020). Greedy decoding had the highest repetition and cross-prompt similarity, while exploratory sampling had the lowest repetition but the weakest lexical and syntactic stability. Conservative top-k sampling produced the strongest overall research-oriented balance, but its average human-review score remained only 2.2 out of 4.

### 4.3 Concrete Generated Behaviors and Failure Cases

For traceability, output IDs in this report use the exact composite identifier `prompt | decoding_setting`, matching the prompt and setting fields retained in the controlled generation artifacts.

Output ID `52.212-4 | greedy` demonstrated severe repetition. The continuation repeatedly returned to the phrase pattern "Contractor shall" and converged toward nearly the same template observed under the other prompts. This is a concrete mode-collapse failure: the model maintained a superficially contractual tone but failed to preserve prompt-specific meaning or produce a coherent clause continuation.

Output ID `52.204-21 | conservative_top_k` had stronger sentence structure and greater variation than the greedy output, but it fabricated or distorted authoritative details such as titles, citations, dates, and obligations. This is the most consequential failure type because the language can appear credible to a reader who does not verify it against the official FAR.

Output ID `52.232-33 | exploratory_top_k` achieved lower repetition and greater character diversity, yet it contained malformed words and broken syntax. The output was less likely to be mistaken for finished legal language, but its reduced coherence also limited research utility.

These cases show why the conservative setting was recommended only for controlled research. It improved structure and diversity relative to greedy decoding, but none of the nine outputs met a threshold for legal, procurement, compliance, or operational use.

# 5. Ethical Considerations and Responsible Use

The central ethical risk is authoritative hallucination: the model can imitate the tone and format of federal contract language without understanding legal meaning or maintaining current regulatory truth. Bender et al. (2021) caution that fluent language-model output can be mistaken for meaningful or reliable language even when it is produced from statistical pattern matching. In this project, that risk is directly visible in the conservative top-k outputs, where plausible structure coexisted with invented or distorted authoritative details.

A second risk is automation bias. A user may give undue weight to a generated clause because it resembles an official FAR provision. The high-stakes context increases the potential impact: incorrect language could affect solicitations, contractual rights, compliance determinations, vendor obligations, or procurement records. Repetition and weak prompt differentiation create a further risk that superficially different requests receive substantially similar boilerplate.

The dataset also represents a frozen regulatory snapshot. Even exact memorization of source language would not guarantee currency after later FAR updates. In addition, prompts containing confidential, procurement-sensitive, personally identifiable, or otherwise restricted information could create privacy and records-management concerns.

## 5.1 Required Controls

1. Label every continuation as synthetic, experimental, and not legally valid.
2. Prohibit direct insertion into solicitations, contracts, amendments, policies, legal opinions, compliance decisions, or procurement records.
3. Verify every generated citation, title, date, threshold, definition, obligation, and clause reference against the current official source.
4. Require review by a qualified procurement, legal, or compliance professional before any substantive use.
5. Retain prompts, random seeds, checkpoints, settings, generated outputs, and review decisions for auditability.
6. Use only within an isolated research or demonstration environment.
7. Do not submit confidential, procurement-sensitive, personally identifiable, or otherwise restricted information.
8. Refresh and revalidate the dataset when the FAR changes.
9. Evaluate qualitative failures in addition to predictive metrics.
10. Prohibit autonomous approval, publication, execution, or downstream decision-making.

# 6. Limitations and Future Improvements

## 6.1 Current Limitations

- The character-level model learns local form and punctuation effectively but has limited semantic understanding.
- The context window is 256 characters, which is short relative to many FAR clauses.
- The corpus is a frozen FAC 2026-01 snapshot and may not reflect later regulatory changes.
- The controlled experiment evaluated only two dropout values and one small architecture.
- Quantitative evaluation emphasizes next-character prediction and surface-level generation measures.

- Human-review scores are based on nine controlled outputs rather than a large blinded study.
- The model has no retrieval mechanism to anchor generation to current official clauses.
- The model cannot distinguish a plausible legal statement from a valid legal statement.

### 6.2 Realistic Future Improvements

1. Evaluate subword tokenization and longer context windows while preserving clause-level split controls.
2. Test additional model capacity and regularization settings using a predeclared validation protocol.
3. Add retrieval from the current official FAR and require citations to retrieved source passages.
4. Constrain generation to extractive summarization, clause comparison, or controlled drafting templates rather than open-ended clause creation.
5. Use broader human evaluation with procurement and legal reviewers, blinded output ordering, and explicit factual-verification criteria.
6. Add automated checks for nonexistent clause numbers, incorrect dates, unsupported thresholds, and inconsistent section references.
7. Implement dataset refresh and regression testing whenever the governing FAR snapshot changes.
8. Measure memorization and source overlap more directly before considering any expanded research use.

The most appropriate future direction is not unrestricted legal drafting. A safer research path would combine retrieval from current official sources, constrained output formats, automatic citation validation, and mandatory expert review. Even with those improvements, deployment would require governance, security, records-management, and accountability controls beyond the scope of this capstone.

## Conclusion

This project demonstrates a complete and reproducible generative-model workflow using a custom PyTorch Transformer and a documented public-sector contract-language corpus. The dropout 0.10 model materially outperformed the dropout 0.30 comparison under controlled validation conditions and transferred consistently to the held-out test partition. Controlled decoding also showed that model-selection metrics and coherent surface form are insufficient evidence of legal reliability. The most structured outputs still fabricated authoritative details, while the other settings exhibited repetition or syntactic corruption. The model is therefore restricted to education, research, and demonstration under strict human oversight.

## References


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